



A spatial model of the relationship between seats and votes[☆]

Stephen Ansolabehere^{a,*}, William Leblanc^b

^a Department of Political Science, Massachusetts Institute of Technology, United States

^b Center for the Study of American Politics, Yale University, United States

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ABSTRACT

The empirical relationship between votes received and seats won in an election is widely used to infer whether district boundaries and other election rules benefit one party or type of voter. Specifically, the electoral system is said to be biased against one party or group if, when that party or group wins 50% of the vote it receives less than 50% of the seats. However, a definitive interpretation of the empirical relationship between seats and votes presents substantial difficulties, which owes much to the absence of the linkage between this relationship and individual voter's preferences and behavior. We offer such a linkage and derive a particular form of it from the one-dimensional spatial model of voting, the most widely used model of voting in Political Science. We show that under this model the methods commonly used to estimate the relationship between seats and votes do not properly identify the electoral bias caused by districting. We use the model to develop a new approach for estimating the electoral bias and apply the techniques to the US Senate during the period 2002–2006 and the US Presidential race of 2004. We find evidence of bias against Democratic voters in the US Senate but not in the Electoral College.

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1. Introduction

There are two ways to view who wins an election—the party that wins a majority of the popular vote and the party that wins a majority of legislative seats or delegates. It is hoped that the party that wins a majority of the vote also wins the majority of the seats, but that is not always the case. Elections in which one party wins a majority in the popular vote but fails to capture a majority of legislative seats or delegates are often viewed as undemocratic and are thought to reflect unfair election laws that systematically advantage one party or type of voter (see [1]).

Perhaps the most common and widely studied problem arises in systems that use plurality rule in legislative districts to elect their representatives, including the United States and the United Kingdom. In plurality rule systems, the nation is divided into distinct geographic units, called districts. The candidate or party that wins a plurality of votes within a given district wins that seat, and the number of seats that a party wins equals the number of districts in which it received a plurality of the vote.¹

For more than a century, political scientists and statisticians have studied the empirical relationship between seats and votes to measure the extent of discrepancies between the division of the popular vote and the division of legislative seats and to detect the conditions when one party might win a majority of the national vote but lose a majority of the seats [2].

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* Corresponding author.

E-mail address: sda@mit.edu (S. Ansolabehere).

¹ The primary alternative is proportional representation, in which the share of seats won by a party equals its share of votes received.

It has long been conjectured and observed that under plurality rule the party that wins a majority of vote will win an even large share of seats [3].² Of greater concern is what political scientists have termed electoral system bias. Electoral bias is defined for an hypothetical election in which each of two competing parties wins half of the national vote. In an unbiased system, each party is expected to win half of the seats when their vote shares equal .5. The bias of the system equals the deviation of the predicted seat share of a party from .5 in an hypothetical election in which the party receives a .5 vote share. “Gerrymandering”, which occurs when those who create the districts manipulate the boundaries, sociological composition, or other features of the election rules in order to gain political advantages, is one important source of electoral bias (see [1], [4]). It is possible, for example, to concentrate a large number of voters of the opposition party into a small number of districts, so that when the parties split the vote evenly, the opposition party wins less than half of the seats [5].

The statistical relationship between seats and vote has since been formalized in two distinct ways. One approach [6] is to regress a party’s share of seats won on its share of the national vote received across several elections. Another approach [7] is to estimate the seats–votes relationship from the cumulative distribution of votes won by a given party across districts in a given year. This method involves adding a small share (say .01) to the share of the vote won by a party in every legislative district (a uniform swing) and, then, calculating how many additional seats the party would win were its share of the national vote to increase by the amount of the uniform swing. The bias is calculated by imposing a uniform swing sufficient to make the national vote share of a party equal to .5 and, then, calculating the share of seats that would be won if the vote share equalled .5.³ Political scientists, election commissions, and courts of law rely on these methods to determine the magnitude of bias in election districts [5].

These methods, we argue, are insufficient to determine whether district boundaries per se create electoral system bias. The relationship between seats and votes depends on the distribution of voter preferences, say from liberal to conservative, in the population as a whole and the distribution of different types of voters across districts. It is the latter that is of interest in determining how votes translate into seats, but, as the model presented here reveals, the two methods described above conflate these two distributions. As a result, using traditional methods, one cannot determine if electoral system bias stems from a skewed distribution of voters across districts or from a skewed distribution of types of voters in the population as a whole.

In order to clarify and gain leverage on this problem we link the seats–votes relationship to the most widely used mathematical model of voting—the one-dimensional spatial model [9]. The one-dimensional spatial model allows us to characterize voter preferences, to define the share of the vote that each party will win, and to determine which party will win which district and the share of total seats. As a result, we can specify the distribution of voter preferences in the population, the distribution of voters across districts, and the seats–votes relationship.

The one-dimensional spatial model assumes that the choices voters face are characterized by a continuous line, commonly called ideology or the ideological dimension. It describes the ideological orientation of government policy from Liberal to Conservative; in the field of public finance, the line is interpreted as the tax rate or the level of government spending. Each voter most prefers a point on that line, called the voter’s “ideal point”, and he or she likes less any point that deviates from that ideal point. Following Downs’ treatment, we formalize preferences as the negative of the distance from the voter’s ideal point. Voters may be ordered from the most liberal to the most conservative, and the distribution of voter preferences is the cumulative distribution of ideal points from most liberal to most conservative.

We assume throughout that there are two, and only two, parties competing in a national election for control of the legislature. Competing parties or candidates each chooses a point, called a policy, on the line that the party or candidate will implement if elected. The share of votes won by a party equals the share of the electorate that is closer to that party than to the competitor. An important concept in the one-dimensional spatial model is the “median voter”. The median voter has an ideal point such that half of the voters have ideal points that are to the left along the line and half are to the right. The median voter is pivotal in the election in the sense that the party closest to the median wins a majority of votes.⁴

We further assume throughout that every legislative district chooses exactly one legislator (single-member districts). Voters are apportioned into districts, and every district contains a distribution of voters from most conservative to most liberal within the district. We further assume that all legislative candidates in a given party promise the same policy. Consequently, the party closer to the ideal point of the median voter in each district wins that seat. We may order districts according to their medians from the district with the most conservative median voter to the district with the most liberal median voter. The cumulative distribution of district medians will be called the distribution of voters across districts.

This model allows us to formulate the relationship between seats and votes. First, define the values of the ideological dimension that correspond to specific vote shares for a party. This may be accomplished by inverting the distribution of voter ideal points (if it is invertible), or calculating the quantiles of the distribution of voter ideal points. Second, the seats–votes curve can be defined as the cumulative distribution of district median ideal points. The median ideal points may be transformed into values of the vote using the inverse of the distribution of voter ideal points. This linkage allows us to express the share of seats a party receives as a function of the share of the vote won nationwide and to develop a new method for estimating the seats–votes relationship.

² It is debatable whether this is a desirable feature of the electoral system. Some political scientists see an advantage to the tendency of plurality rule to inflate the majority party’s seat share to be higher than its vote share.

³ Gelman and King [8] modify the uniform swing model above to account for incumbency and allow for different district variances.

⁴ If the parties are at the same point, all voters are indifferent and are assumed to toss an unbiased coin to choose one of them. These assumptions are standard in the one-dimensional spatial model. See [9].

Section 2 of this paper presents the basic model; Section 3 discusses empirical estimation methods. The final section of the paper presents an application to the Electoral College.

2. Framework and model

In this section we construct a model that adapts the one-dimensional spatial voting model to the relationship between seats and votes.

2.1. Parties and voters

Let there be two parties, labeled X and Y , and a real line that corresponds to a continuum of ideologies or possible policies. Each party is associated with a policy, x and y respectively. We will assume throughout $x < y$, i.e., party X has a policy distinct from and to the left of party Y 's policy along the ideological dimension. For now we shall treat x and y as constant over time, although we will relax this assumption below.

Let the ideal point of a given voter i be given by w_i .⁵ The fraction of voters with an ideal point less than a given value z is given by the continuously differentiable cumulative distribution function $F(z)$. This function will be discussed in more detail below. Let the real number v_t describe a time-varying net “valence” preference each and every voter has for party X . v_t can be negative, indicating a net preference for party Y . Valence can be thought of as a general party reputation for competence, how well the current party in power managed the economy, and other general swings in support for one party over the other.

We assume that voter preferences can be summarized by a “utility” function. Each voter supports whichever party gives the voter more utility. Let utility for voter i for supporting party X be $u(x - w_i) + v_t$ with $u(z)$ twice continuously differentiable, strictly concave, and with a unique maximum at 0. Therefore, a voter has a stronger preference for party X if the party's policy is closer to her ideal point and if v_t is large. Let voter utility for party Y be $u(y - w_i)$. Assume voters vote for the party from which they get higher utility in any given election. (That is, there is no abstention, attempts to manipulate future party policies, and so forth.) In terms of the utility function, a voter supports party X if $u(x - w_i) - u(y - w_i) + v_t > 0$ and party Y if $u(x - w_i) - u(y - w_i) + v_t < 0$.⁶

Of particular importance is a voter who is indifferent between the two parties. Any voter for whom $u(x - w_i^*) - u(y - w_i^*) + v_t = 0$ is indifferent between the two parties, and the point w_i^* is the ideal point of the indifferent voter, given the policies of the parties and the value of the valence term. Because we have assumed strict concavity and $x < y$, $u(x - w_i) - u(y - w_i) + v_t$ is strictly decreasing in w_i .⁷ That is, as a voter becomes more conservative, holding party policies and valence fixed, the voter's net preference for party X decreases. Therefore, there is at most one solution w^* to the equation $u(x - w^*) - u(y - w^*) + v_t = 0$.⁸ If such a point exists, we will call it the “cut-point” and label it c_t . Furthermore, if a voter is more conservative than the cut-point, that voter will support party Y and more liberal voters will support party X . For example, for quadratic $u(z)$, $u(z) = -(z)^2$, the cut-point is $c_t = \frac{x+y}{2} + \frac{v_t}{2(y-x)}$.

Note, the purpose of the spatial component of the model is to generate the concept of a cut-point. The cut-point implicitly depends on party policies and valence. Up to this point, we have assumed valence can vary but party policies are fixed. We could have just as easily had a fixed valence and variable party policies, or allowed both to vary. Many alternate models that also imply there being a single variable which divides voters into supporters of one party or another would work for the analysis that follows.⁹ For example, suppose voters have a “partisan intensity” for party X , q_i . This term could be negative, indicating a tendency to vote for party Y . Suppose further, like in the spatial model, there is a universal valence factor, v_t , indicating an election-specific tendency for all voters to vote for party X . We can then define a voter utility as $q_i + v_t$. Voters with q_i such that $q_i + v_t > 0$ vote for party X ; voters with $q_i + v_t < 0$ vote for party Y . Therefore the cut-point is $-v_t$.

⁵ For now we will treat policies and ideal points as constant over time. These assumptions will be relaxed below.

⁶ Nominally utility functions do not need to be symmetric. However, one could always transform coordinates to make utility functions symmetric. Such a transformation will change both the distribution of voters and districts which will be discussed below. Essentially, one can without loss of generality impose symmetry on the utility functions and then study the symmetry or lack thereof of the distribution of voters and districts.

⁷ How $u(x - w_i) - u(y - w_i) + v_t$ changes with changes in w_i is given by its derivative with respect to w_i , $-u'(x - w_i) + u'(y - w_i)$. Since $x < y$, $x - w_i < y - w_i$. Since $u'' < 0$, $u'(y - w_i) < u'(x - w_i)$. Therefore, $-u'(x - w_i) + u'(y - w_i) < 0$. Therefore net utility for left party is indeed decreasing in conservatism.

⁸ Because u is continuous, if there is at least one w_i such that $u(x - w_i^*) - u(y - w_i^*) + v_t > 0$ and at least one w_k such that $u(x - w_k^*) - u(y - w_k^*) + v_t < 0$, then the solution will exist by the intermediate value theorem. Therefore, if there is no solution for a given v_t then it must be the case that every voter supports one party or the other, and vote share and seat share will be equal to 100% for that party.

⁹ The model can also be generalized beyond cut-point models, as long as seats and votes are both increasing functions of the same single underlying variable. For one example, [10] the model where vote share depends on a random median voter could be adapted for this framework. For another example, suppose each voter supported party X with a certain probability that depended on valence, $P_i(v_t)$ with $P_i(v_t)$ increasing in v_t for all i and strictly increasing for at least one i for all v_t . Then vote share $F(v_t)$ will be average $P_i(v_t)$ (which is strictly increasing). District vote shares would be average vote share in each district. $H(v_t)$ is then the number of districts with at least 0.5 vote share, which is non-decreasing in v_t . Equating $v_t = c_t$ allows one to proceed with most of the rest of the analysis of this paper as normal. Of course, the exact interpretations of F and H change, and the process to find F may change as well based on the underlying model.

Partisan intensity or loyalty may be incorporated in the model by equating $q_i = u(x - w_i) - u(y - w_i)$.¹⁰ Of course, the exact interpretation of the results depends on the underlying model. For this paper, we will use the ideological voter interpretation.¹¹

2.2. Vote share and seats

Given the one-dimensional spatial model of voting that implies a cut-point, c , we can now analyze the seats–votes relationship. Recall that $F(z)$ is the cumulative distribution function describing the fraction of voters with ideal points $w_i < z$. F is defined with respect to an electorate. There may be one F for presidential voters, another F for average district votes, separate F 's for each district, and so forth. We will write aggregate (as opposed to individual district-level) cumulative distribution functions as F , without a subscript. We will write district-level vote shares as F_j where j designates the district. For both aggregate and district-level distributions, assume F and all F_j are continuously differentiable with derivative $f(z)$ and $f_j(z)$ (which we will at times refer to as the “density” of voters) and inverse $F^{-1}(\theta)$ and $F_j^{-1}(\theta)$. Furthermore assume there exists an interval on the real line such that $f(z) > 0$ and $f_j(z) > 0$ on the interval. With this assumption we can ignore the choice of voters with ideal points exactly at the cut-point when we restrict attention to cut-points within the interval, since such voters constitute measure zero. By the definition of F , the fraction of voters, V , voting for party X is then $F(c)$, the cumulative distribution of voter ideal points evaluated at the cut-point.

For party X to win a majority of seats in any given district, j , the party must win a majority of votes from that district. To do this, the median voter of the district, $m_j = F_j^{-1}(0.5)$, must have an ideal point left of the cut-point. In terms of notation, district j supports party X if $m_j < c$ and party Y if $m_j > c$. We will ignore tied districts. Therefore to characterize district behavior we only need to know the preferences of the median voter in each district. Note this result relies heavily on the assumption that voter behavior depends only on a one-dimensional variable. That is, there are no district- or candidate-specific effects, such as incumbency. In Section 3.4 we will consider an extension accounting for incumbency effects.

One can generalize the concept of seats won to study presidential races by thinking of states as districts. The number of “seats” a presidential candidate wins is equal to the number of electoral votes secured. Just as in the legislative district case, a candidate will win a state and the corresponding electors if the median voter of the state is left of the cut-point.

Let $H(z)$ be a cumulative distribution function describing the fraction of districts with district median voters such that $m_j < z$. The seat share party X wins, S , is then given by $H(c)$.

Up to this point we have treated party policies, voter ideal points, and identities as fixed over time. To model possible changes, one can simply think of F and H changing with time.

If voters are held constant in the general population, but district boundaries change, F must remain constant but H might change. In this sense H represents district boundaries. However, it is possible for H to change if district borders remain fixed but voters move from one district to another. One might argue such a change represents a kind of passive, natural redistricting. Furthermore legislatures might choose not to redistrict at all and allow natural redistricting to do gerrymandering for them. It may also be possible to redraw borders without changing H .¹²

Such a change will not alter the seats–votes relation, as will be shown below. Therefore, it is unclear whether such a change would have any political impact.¹³

2.3. Seats–votes relationship

We now have expressions for the votes for party X and the seats of party X in terms of one variable, c . Let the function G be defined as the relationship between seats and votes $S = G(V)$. Because F is invertible, we have the very simple but powerful relation between seats and votes:

$$S = G(V) = H(F^{-1}(V)).$$

The seats–votes curve is therefore made up of two parts—the distribution of voters’ ideal points in the general population, F , and the distribution of the ideal points of the median voters across districts, H . As noted above, H can be thought of as representing the political impact of district boundaries.

Concerns about electoral system biases introduced by districting are reflected in the distribution H . Unfortunately, using just information about seat shares (S) and vote shares (V), one generally cannot estimate the distribution function H separately from the distribution function F .

¹⁰ See [11] for an example where voter preferences change with policies. Such a model would be difficult to merge with the model here.

¹¹ Considering other interpretations of F may very well be fruitful, but are left to future work and are beyond the scope of this paper.

¹² For example, consider two adjacent districts with district medians m_1 and m_2 . Move two voters from district one to district two with ideal points $z_{1a} < z_{1b}$, and 2 voters from district two to district one with ideal points $z_{2a} < z_{2b}$. If $z_{1a} < m_1 < z_{1b}$, $z_{1a} < m_2 < z_{2a}$, $z_{2a} < m_1 < z_{2b}$, and $z_{2a} < m_2 < z_{2b}$ then the new district medians will not change.

¹³ Such a result depends on the assumption that there is no incumbency effect, or at least incumbency is not influenced by neutral gerrymanders. If incumbents build long-run constituencies that get disrupted by changing borders, even by a neutral gerrymander, then election outcomes will be impacted. See [12] and [13]. Such issues are beyond the scope of this paper.

A stylized example demonstrates this point more concretely. Suppose voter ideal points are uniformly distributed from $-a$ to a with $a > 0$. This corresponds to a cumulative distribution function $F(z) = 0$ for $z \leq -a$, $F(z) = \frac{z+a}{2a}$ for $-a < z < a$ and $F(z) = 1$ for $z \geq a$. The corresponding inverse is $F^{-1}(\theta) = 2a\theta - a = a(2\theta - 1)$. Suppose further that district medians are uniformly distributed from $-b$ to b with $b < a$. The corresponding cumulative distribution function is given by $H(z) = 0$ for $z \leq -b$, $H(z) = \frac{z+b}{2b}$ for $-b < z < b$ and $H(z) = 1$ for $z \geq b$. The seats–votes relation $S = G(V) = H(F^{-1}(V))$ is therefore $S = 0$ if $a(2V - 1) < -b$ or equivalently $V < \frac{1}{2}(1 - \frac{b}{a})$, $S = 1$ if $V > \frac{1}{2}(1 + \frac{b}{a})$, and $S = \frac{a}{b}(V - \frac{1}{2}) + \frac{1}{2}$ otherwise. That is seat share is a linear function of vote share in this example. A linear relationship is approximately consistent with many observed seats–votes relationships; see [1].¹⁴

Note the seats–votes relationship is determined by the ratio of the parameters of a and b . In other words, one can change the distribution of district medians, represented by parameter b , without changing the relationship between seats and votes, as long as the corresponding distribution of voter ideal points also changes so that the ratio $\frac{a}{b}$ remains constant. Since changes of b represent gerrymanders, a gerrymander could be rendered invisible to seats–votes studies if the distribution of voters in the aggregate changes simultaneously.

Even though the gerrymander is invisible to traditional seats–votes analysis, it can still have real political impact. Suppose the liberal party has a slight valence advantage, so the cut-point is slightly right of center $c > 0$. The seat share the liberal party wins is therefore $H(c) = \frac{c+b}{2b}$ or 100% if $c > b$. That is liberal party seat share is strictly decreasing in b if $c < b$ (and if $c > b$ then reducing b does not hurt the liberal party.) If the liberal party has a valence advantage they would be helped in terms of seat share by a gerrymander that reduces b , regardless of changes in a .

It is worth emphasizing that the analysis depends on the existence of a single cut-point that applies to all districts. In other words, all candidates from a given party announce the same policy and are subject to the same (national) effects on the vote. If one were to allow candidate- or district-specific effects, then the pivotal voter in each district would no longer necessarily be the median voter. One could, however, analyze the distribution of pivotal voters instead of the distribution of district medians. The logic of the central result of the model will still apply, in the sense that the seats–votes relation is a mixture of two distributions: the distribution of voters in the aggregate and the distribution of pivotal voters across districts.

3. Estimation methods

The spatial model suggests a two-step approach for estimating the seats–votes relationship. First, estimate the distribution of voters' ideal points in the electorate as a whole (i.e., F). Second, use the estimate of F and data on national shares of votes and seats won by a party to estimate H .

3.1. Finding F

There is a long and robust literature on measuring ideology among voters. At least three different types of data are commonly used:

1. Survey or exit poll data with direct questions on ideology or partisanship [14].
2. Survey or exit poll data with a battery of questions on various issues. Use the questions to construct a measure of ideology (e.g. factor scores) [15].
3. Find the distribution of demographics from one data source (such as the census). Use survey data to find a mapping from demographics to ideology and/or voting behavior. Combine the two sources to find distribution of ideology.

Which way of estimating F is preferred depends on the availability of data and its quality. In general, we prefer surveys with very large numbers of cases (say, larger than 10,000) and with direct measures of ideology, along the lines of option 1. Some survey data will be too grainy to estimate F smoothly. Even with grainy data one can still approximate another useful item: the density f . For example, one might only know voter ideologies on a three-point scale (liberal, moderate, conservative.) For this case we can still construct the measure: approximate f with the number of moderates divided by total number of voters. Alternatively, one could use the fraction of people near median income, or the fraction of people who self-identify as independents with no leaning toward either party.

In addition, some care must be taken in the exact interpretation of scale of the observed variable because a unit change in the cut-point will depend on the measure used. For example, if one uses linear factor scores derived from questions on policy, a one unit increase in conservatism represents changes in a subject's probability of giving more conservative answers to policy questions. If one chooses a metric without a scale interpretation (e.g. raw number of conservative answers to questions) then constructs derived from ideology (like H) may be difficult to interpret in absolute terms. However, comparative statements (such as "the median state is more conservative than the median House district") should still be possible.

Use of demographic indicators, as in approach 3, is probably the least desirable as that is the least direct indication of voters' ideologies. That said, there may arise problems in which the dimension of interest is the demographic factor, in which case the demographic indicator is preferred. For example, if one were interested in how income inequality was reflected in representation, one might want to construct a seats–votes relationship as a function of the income distribution in the society.

¹⁴ One could reasonably argue that the distribution of district medians should not be continuous, since there are a finite number of districts. This case is only meant as an example. Examples with non-continuous distributions of district medians are also possible.

3.2. Finding G and H with district-level data

Finding H with full information of district-level ideology data is simple. Pick an ideology, z . Then $H(z)$ is the number of districts such that $F_j(z) > 0.5$. Once one knows H and F then one can simply use the formula $G = H(F^{-1}(V))$ to find G .

An equivalent way to view the above procedure is to consider the median voter of each district, $m_j = F_j^{-1}(0.5)$. The median voter is more conservative than the $F(m_j)$ share of the population. For example, suppose the median voter of a district is more conservative than only 40% of the country. That is, the district is a relatively “liberal district”. That district will support the left party if the overall vote share of the left party is at least 40%. If one knows the $F(m_j)$ of the median voters of each district, one can count the number of district medians more liberal than a given vote share to construct the model’s prediction of the seat share for any given vote share.

However, finding the median voter for every single district may be difficult. If one has information for $f_j(c_0)$ we propose using a modified version of a technique known as uniform swing [7]. The uniform swing method has the vote share in every district increase by one unit when the aggregate vote share increases one unit. In terms of the model, the uniform swing hypothesis amounts to assuming that for each district there exists a constant initial value a_j such that $F_j(c) = F(c) + a_j$ and $f_j(c) = \kappa$ with κ constant.¹⁵

A natural extension to the uniform swing hypothesis is to allow $f_j(c)$ to vary with j but remain constant with respect to the cut-point (at least for small deviations from an initial cut-point). Let V_{0j} be the actual vote share in district j for the election under consideration. Let c_0 be the cut-point in an open seat election. Then the district vote shares are $F_j(c_0 + \delta) \approx f_j(c_0) * \delta + V_{0j}$. This approximation allows one to find H , as above. Let V_0 be the actual aggregate vote share for the election under consideration. The aggregate vote share is $F(c_0 + \delta) \approx f(c_0) * \delta + V_0$. With this approximation one can find G as above.

One can also find G directly by thinking of the process as a uniform swing with weights. We will refer to this technique as the “weighted swing method”. When the aggregate vote share changes by one unit, each district will change $f_j(c_0)/f(c_0)$ units.¹⁶ Formally, by substituting out δ above, $F_j(c_0 + \delta) \approx \frac{f_j(c_0)}{f(c_0)} * (F(c_0 + \delta) - V_0) + V_{0j}$. In terms of intuition, some districts have more moderates than others. A district with a large number of moderates will change vote share more rapidly than districts made up of extreme, intransigent partisans.

It is worth emphasizing that the weighted swing method relies on the local approximation of the density of voters within a district. As one considers vote shares further and further away from the actual election outcome, the estimates will rely more on the assumptions and less on the data. Most standard approaches in the literature, such as uniform swing and Judgelt [8], are subject to the exact same critique. In contrast, measuring median voters directly does not rely on the linearity assumption.¹⁷ In Section 4 below, we will compare uniform swing, weighted uniform swing, and a direct estimate of median voter methods to each other for estimating H . Each method relies on different data sources (uniform swing—election data only; median estimate—survey only; weighted swing—hybrid) and different theoretical assumptions. To the extent the different approaches produce similar estimates of the seats–votes relationship, the more confidence one can reasonably have in the evidence concerning whether there is electoral bias and which party it favors.

3.3. Finding G and H with aggregate data

The other standard approach to finding the seats–votes relationship is to consider multiple elections and regress seats on votes.¹⁸ A tacit assumption behind this approach holds that the seats–votes relationship is constant over time. The regression method also ignores district-level information; there is only one effective observation per election, and that is the national result.

Regression models of the seats–votes relationship often take non-linear forms. Kendall and Stuart [3] developed the law of Cubic Proportions, which states that the odds of winning a seat equal the cube of the odds of winning a vote: $\frac{S}{1-S} = (\frac{V}{1-V})^3$, which approximates the cumulative normal. More generally, this is parameterized as: $\frac{S}{1-S} = A(\frac{V}{1-V})^\beta$, where $\ln(V)$ is the residual, $\ln(A)$ is the intercept, and β the coefficient on $\ln(V/(1-V))$ from the regression of $\ln(S/(1-S))$ on $\ln(V/(1-V))$. Cox and Katz [5] applied this model to US House and state legislative elections and used it to address questions about the fairness of elections overall and of the various agencies that draw districts, such as courts, commissions, and the state legislatures.

¹⁵ Note, the formula makes it very clear that the uniform swing assumption becomes less reasonable for large vote deviations from the actual election; the method yields physically absurd predictions (district vote shares over 100%) for large vote deviations.

¹⁶ This technique is similar in spirit to [8]. However, [8] estimates how much each district should change based on past behavior of the district; this technique uses outside information plus a theory of voters to estimate how much each district changes.

¹⁷ However, the direct median voter measure relies heavily on the survey being unbiased, and comparing the measure to the uniform and weighted swing approaches depends on the assumptions of the model. In particular, comparisons depend on the absence of district-specific effects that have not been explicitly accounted for.

¹⁸ By regression we mean ordinary least-squares linear regression, non-linear regression, non-parametric regression, etc. In practice in the literature this often means linear regression or fitting a particular functional form (such as the logistic). However, the discussion here can be applied to a wide variety of techniques.

Under the model described in the previous section, the seats–votes function G is a composite of two different functions, H and F . One should therefore be less likely to suspect that G will be a simple linear, symmetric function. Even if one had reason to believe one of H or F was a simple function, the alternate might not be simple, leading to a complex G . Hence, the analysis here points to the use of more functional forms of G that allow for non-linearity. For example, instead of regressing S on V , regress S on V , V^2 , and V^3 , as in [6], to estimate G .

Distinguishing H and F in the regression model, however, is not possible. The regression method posits a relationship between seats and votes over time $S_t = G(V_t) + \epsilon_t$ with ϵ_t independent of V_t . Plugging in for G , we have a new regression model $S_t = H(F_t^{-1}(V_t)) + \epsilon_t$. The parameters of H and F are not separately identified in this model.

One strategy for salvaging the regression model is to use survey or other data to characterize the distribution of voter preferences. Specifically, first estimate $F_t^{-1}(V_t)$ from survey data and use the estimate of F^{-1} to transform V . Then use the transformed vote, $\hat{F}_t^{-1}(V_t)$, as the independent variable in a non-linear regression in which S_t is the dependent variable.¹⁹

3.4. Accounting for incumbency

The above approaches have assumed that voters vote solely based on their own individual characteristics, and that there are no substantial district- or candidate-specific effects. This section will describe how to control for one particular candidate-specific effect, incumbency. The approach described here could be generalized to any observable district traits that are hypothesized to impact vote percentage or voter utility.

One blanket solution to the incumbency effect is to use an alternative measure of vote share that does not depend on district incumbency, such as the party's share of the presidential vote in the district. In this case, a seat would be counted as won if it receives more than 0.5 presidential support. Assuming the alternate measure is truly independent of district incumbency, this should be a perfectly adequate way of finding H . However, it may be troublesome to interpret this as a measure of “the” seats–votes curve, because it is a different object, relating a different measure of seats to a different measure of votes.

A standard approach for those estimating seats–votes relationships involves estimating the incumbency effect and adjusting the observed vote. To estimate the incumbency effect regress district vote share on an indicator for incumbency and a measure of the underlying (or “normal”) party division, such as lagged vote or presidential vote in the district [16,17]. One can then add in/subtract off the incumbency effect as needed [8]. In our model, let the vote share in a district be a mix of an ideological component and an added incumbency term $V_j(c, I_j) = F_j(c) + \gamma * I_j$ with $I_j = 1$ for an incumbent of party X , $I_j = 0$ for an open seat, and $I_j = -1$ for an incumbent of party Y .

For the district-level analysis using full information of F_i above, incumbency distorts the estimation of the seats–votes curve but not H . H is directly found from the full knowledge of the distribution of voter ideology. The hypothetical seats–votes relationships that are constructed, however, assume there is no incumbency effect. To find the seats–votes curve one must add the estimated incumbency effect back into the district vote share for the incumbent.

For the weighted swing method, the incumbency effect is already accounted for by the initial district values. One has in a sense the opposite problem. The weighted swing provides an appropriate estimate of the seats–votes curve without any adjustment for incumbency, but to estimate H properly one must subtract the incumbency effect out district by district. This will yield a new set of district vote shares. Count the number of districts with modified vote share over 0.5 to find the estimate of H .

Incumbency adjustments present an immediate problem for regression methods, both in the context of our model and more broadly. The number and distribution of incumbents, which change from election to election, influence the seats–votes relation. Therefore, in general there will be no unique seats–votes relationship. This is a similar problem to the case where F changes over time. More troubling, the percent of incumbents representing a given party in an election depends strongly on how well that party did in the prior election. This creates a serious problem for regressions of seat shares on vote shares over time methods, because these methods assume either there is a unique seat share associated with each vote share, or at least any deviations from the expected seat share for each vote share is independent.

Inclusion of district and candidate-specific factors present an area for further extensions of the model and empirical methods.²⁰

4. Example

In this section we present results of applying several of the methods described above to find seats–votes curves without incumbency effect for the year 2006 (using survey data and Senate data) and 2004 (using presidential data.) First, we will

¹⁹ As mentioned above, the standard regression approach implicitly assumes a constant seats–votes relationship. The model makes it clear that this assumption could be violated either by H changing or F changing. The ideological voter model gives the option to relax the assumption of constant F . Maintaining the assumption of constant H , find H as described in the above paragraph. G_t is then given by G_t with $G_t = H(F_t^{-1}(V))$.

²⁰ The spatial model points to another approach. Suppose $V_j(c, I_j) = F_j(c + \gamma * I_j)$ with I_j an indicator variable for incumbency status. Suppose one has full information on district-level ideology, $F_j(c)$. Divide districts into open seats, seats with party X incumbents, and seats with party Y incumbents. These groups define three different group aggregate vote shares, F_0 , F_X , and F_Y . Each of these subgroups define an initial cut-point $F_0^{-1}(V_0)$, $F_X^{-1}(V_X)$, $F_Y^{-1}(V_Y)$. With these three cut-points one can determine which districts will vote for which party using $F_i(c)$. To consider alternate vote shares, one can alter the cut-points (holding the differences between the group cut-points constant).

directly measure the median voter of states using survey data. Second, we will use state-by-state vote share for Democratic presidential candidate John Kerry in the 2004 general Presidential race.²¹ We also present predictions concerning the Electoral College, to see whether Kerry would have won 50% of the electors (and thus the Presidency) if he had won 50% of the vote. Finally, we will use Senate election results for 2002–2006, applying the weighted swing and uniform swing methods.

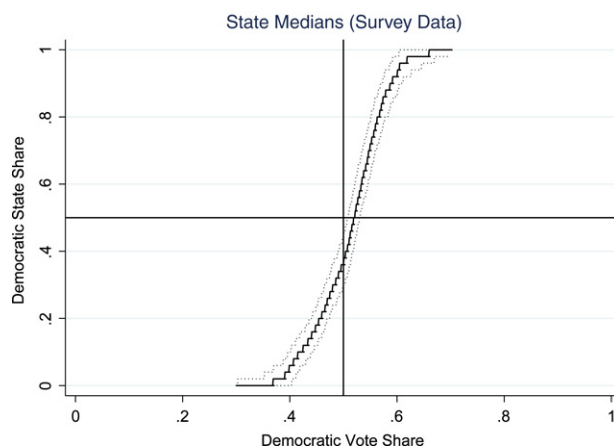
The uniform swing model roughly approximates the weighted swing model, although it deviates more from the theoretical graph for vote shares around 0.6. In this example, the uniform swing method is arguably still a reasonably good approximation for seats–votes. The aggregated curves, especially using the weighted swing model, do mostly lie within the confidence intervals of the survey based seats–votes prediction. In particular, each of the methods shows that at 50% Democratic vote share, Democratic seat share is less than 50%. This is known as “bias” in the seats–votes literature and generally taken to be normatively troublesome. For example, Tufté says [1] “. . . electoral systems biased toward a particular party are hardly defensible, at least in relatively democratic systems . . .” The degree of bias measured varies based on the method, but the presence of at least some bias is robust.

4.1. Direct measure

We used the Cooperative Congressional Election Survey to construct direct measures of voter ideology state by state. The survey has 36,051 subjects, with respondents from all 50 states and the District of Columbia. The survey included several dozen policy questions, which we used to generate principal factor scores for each subject.²² The topics these questions deal with include increasing the minimum wage, ending the Iraq war, amnesty for immigrants, stem cell research, banning “partial birth” abortions, environment, social security, capital gains tax, union influence, tax rates (two questions), and use of military (for defense of oil supply, spreading democracy, destruction of terrorist camps, defense of allies, and helping the UN uphold international law).²³

As discussed in Section 3.2 above, to find the hypothetical seats–votes curve, pick a vote share. Then find the ideology of the voter such that V voters are more liberal, $c = F^{-1}(V)$. The corresponding seat share is the number of states with median voters left of c , $H(c)$. An easy way to do this is to work with transformed scores instead of the raw principal factor scores. Transform scores so that $z' = F(z)$. The most liberal voter has a new score of 0; the most conservative voter has a new score of 1; the median voter in the population as a whole has a score of 0.5; the voter with a score of 0.75 is more conservative than 75% of the population, and so forth. Then $z' = F(z) = F^{-1}(z') = V$ by construction.

However, despite the large number of survey respondents, there is still the possibility of substantial survey measurement error. To account for this we bootstrapped confidence intervals by re-sampling the population of survey respondents 1000 times. For each re-sample we calculated a seat share for each vote share. The following graph shows the 25th, 500th, and 975th seat share for each vote share. The middle line therefore represents a point estimate for the hypothetical seats–votes curve, while the dotted lines represent an estimate of the 95% confidence interval.



²¹ In this section third-party votes are dropped from the calculation of Democratic vote share. For example, if both the Republican and Democratic candidates win the same number of votes, their vote shares will both be 50%.

²² Many subjects only answered some questions. Missing responses were imputed as follows. Let answer for person i , question j be y_{ij} . Then $y_{ij} = b_j * x_i + e_{ij}$ with b_j a question specific regression coefficient, x_i a person-specific score, and let e_{ij} be the residual. Estimate b_j using principal factor analysis on those subjects that answered every question. Then for all individuals estimate $x_i = \sum \frac{(b_j * y_{ij})}{b_j * b_j}$, using only available questions. By construction estimated scores for individuals who answer all questions will be the same as the original principal factor scores.

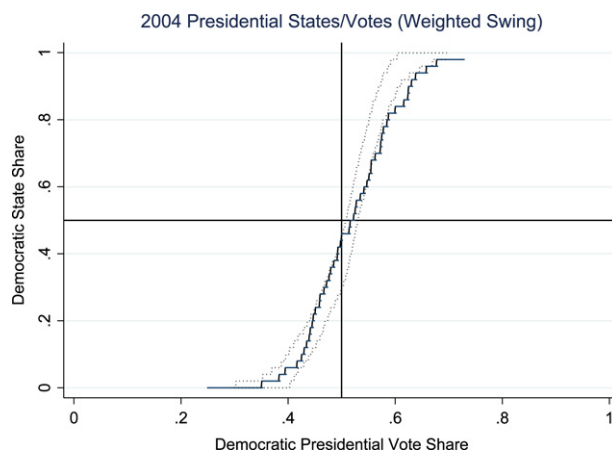
²³ The exact wording for most of the relevant survey questions can be found here: http://web.mit.edu/polisci/portl/cces/material/CCES_Common_Content_August_15_2006_final.pdf. Questions used were 14–18 and 28–33 on the pre-election survey and 16 and 17 of the post-election survey.

At 50% vote share, Democrats win 36% of the seats according to the direct measure, with an upper confidence bound at 44% and a lower bound at 30%. The 36% value however is a bit of a knife-edge case; at 50.1% Democratic vote the point estimate for Democratic seats jumps to 38%. Note that the direct measure only yields percentages divisible by 2 because it assumes both Senate seats switch party support together. Nevertheless, the Democrats winning only about 38% of the seats with a majority of the votes represents a massive measured bias against the Democrats.

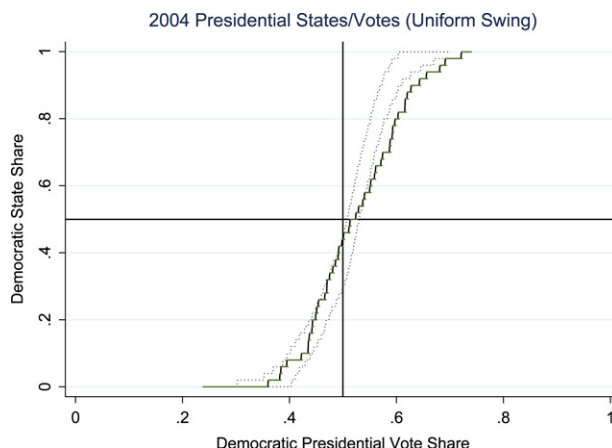
4.2. Presidential vote and states

Here we apply the uniform swing method and the weighted swing method described in Section 3.2 above to the US general presidential election of 2004. In this context, the use of the weighted swing method requires a measure of the number of centrists in the national electorate and in each state. For this example, we used the fraction of surveyed voters within a state with ideology scores in the middle 20%. This measure varied from 13.8% in Washington state to 32.5% in West Virginia. To find states/votes relationship, we begin with the actual vote share in each state and the actual overall vote share in the country. We then deviate each state by a unit step multiplied by the number of centrists in each state.²⁴ The new aggregate Democratic vote is calculated by multiplying the state vote share by the number of voters who voted either Democratic or Republican in the 2004 Presidential election. The Democratic vote share is then the aggregate vote share divided by the total number of Presidential voters. The number of states with Democratic support over 50% is a seat share value. These two values then determine a new seats–votes pair.

The following graph plots the weighted swing states/vote curve as a solid line. The dotted lines represent the 95% confidence interval for the distribution of district medians described in the previous sub-section. If the model is a good approximation for voter behavior and there were no systematic bias from the survey, then the states/vote curve generated from the presidential election data should correspond to the theoretical seats–votes curve generated from the survey responses.



The next graph plots seats–votes curve generated by the uniform swing method (all weights set equal to 1).



²⁴ The step size used was such that if all weights were equal to 1 (the uniform swing method) the vote change would be 0.1%. The number of centrists per state was normalized so that the average weight was equal to one.

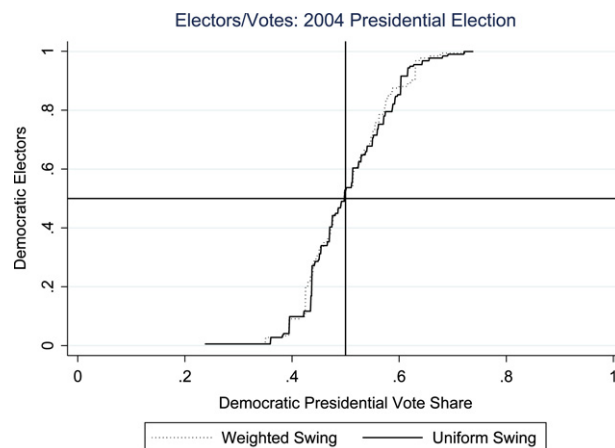
The uniform swing model roughly approximates the weighted swing model, although it deviates more from the theoretical graph for vote shares around 0.6. In this example, the uniform swing method is arguably still a reasonably good approximation for seats–votes.

The fit is not quite perfect. Democratic state wins are measured somewhat high when Democratic vote share is under 50% and somewhat low over 50% relative to the survey model. There are many possible reasons for this. The survey is taken in 2006; the election was conducted in 2004. The model also makes a variety of simplifying assumptions. The model does not take into account campaign effects. Swing states may also be subject to aggressive advertising or media effects, changing voter behavior. The model does not account for turnout effects if turnout changes voter propensity to vote Democratic. In addition to model approximation errors, there is the possibility of survey bias. Survey bias would be especially problematic if the bias is concentrated in swing states.

Despite all these possible problems the presidential states/votes curves (especially using the weighted swing model) do mostly lie within the confidence intervals of the survey based seats–votes prediction. In particular, each of the methods shows a bias against the Democratic party.

4.3. Presidential vote and electors

The following graph shows the relationship between electoral votes and vote share, using both the weighted and uniform swing models.²⁵



The two models give very similar predictions. It is possible that for studying the electoral college, the simple uniform swing method is a “good enough” approximation if a measure of centrists is not available. Interestingly, if Democrats had won 50% of the vote, they would have won a majority of electors according to the point predictions. The Democrats would have had to win about 49.8% of the vote to win a majority of the electors in the weighted swing model; about 49.7% in the uniform swing model. This is some evidence that there is no structural bias in favor of Republican presidential candidates (in the sense that Republicans can win the presidency without winning the popular vote). If anything, the bias might favor Democrats. Note, however, that these estimates ignore the impact of third parties. We therefore speculate that Bush’s ability to win the presidency despite a loss in the popular vote in 2000 was due to third-party influence or idiosyncratic features of the 2000 election (especially in Florida), and not because of a long-term maldistribution of voters across states.

4.4. US Senate vote and seats

The US Senate elections from 2002 to 2006 allow us to examine similar districts to the presidential election of 2004 (i.e., the states) but with the additional complication of incumbency. In order to pool the three elections (2002, 2004, and 2006) we normalized the data to correct for year-to-year fluctuations in the national share of the vote won by Democrats and for the added share of the vote won by incumbents.²⁶ For the uniform swing method, we regressed Senate Democratic vote share on

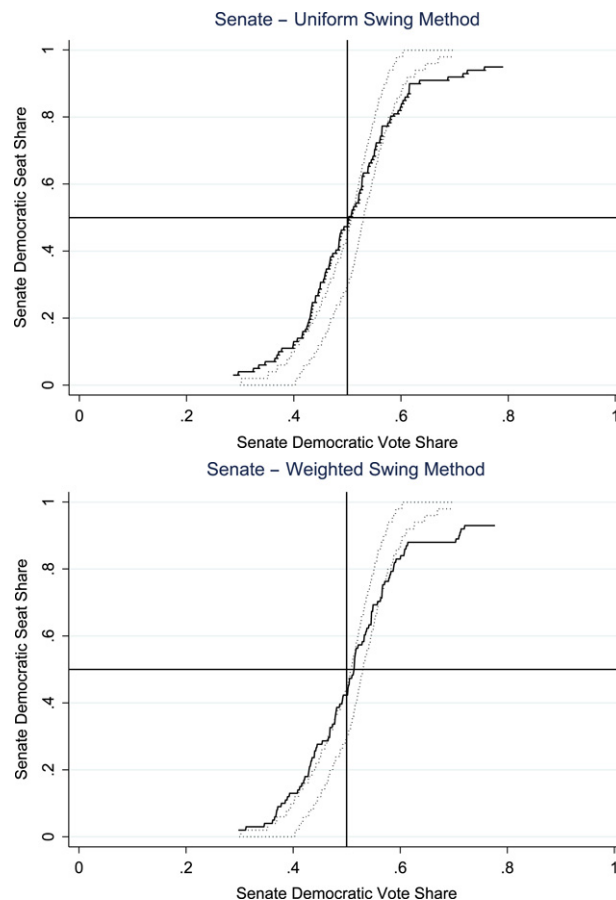
²⁵ The faithless elector in Minnesota who voted for John Edwards is counted as an elector for Kerry in this analysis. Furthermore, states are assumed to cast all electors for the candidate who wins a majority of the vote in the state. That is, we are ignoring the special rules of Maine and Nebraska which assign some electors based on who won a majority in House districts.

²⁶ There were several other quirks with the dataset that had to be dealt with. There were a total of 101 races instead of 100, because Missouri had a special election in 2002. We treated each Missouri race as two-thirds of a seat so that the number of seats added up to 100. Bernard Sanders of Vermont is coded as a Democrat for this analysis. Joseph Lieberman’s victory in 2006 in Connecticut is coded as a Democratic uncontested win. Multiple Democrats ran in the Louisiana 2004 general election; their vote shares were pooled to calculate the total Democratic vote share.

incumbency, year “dummies”,²⁷ a variable equal to one for for 2002 and 2004, and Democratic presidential vote share in 2004 (as a measure of normal vote). The incumbency effect was estimated at 10.4%.²⁸ The 2004 dummy was -3.4% and the 2002 dummy was -6.0% . Since 2006 was a good Democratic year the signs are as expected. We then subtracted off the incumbency effect per district (as per the constant incumbency effect treatment described in Section 3.4) and added in the year-specific effects to transform the Democratic vote share to look like an open election in 2006. For example, if the Democrats won 40% of the vote when facing a Republican incumbent in 2002, that district would be treated as $40 + 10.4 + 6 = 56.4\%$ Democratic vote share.

For the weighted swing analysis, we weighted incumbency and year effects by the number of centrists in a state. That is, the year and incumbency effects varied state by state depending on the number of centrists in that state.²⁹

Uncontested elections were dropped from the incumbency analysis and treated as either always Democratic or always Republican wins. Therefore, in the following graphs at extreme vote share values Democratic seats do not converge to zero or one. However, recall that both the uniform swing and weighted swing methods rely on local approximation arguments, and should be considered unreliable at extreme deviations from the actual election anyway. Uncontested races were also dropped from the determination of average overall vote share.



As with the Presidential case, the weighted swing method corresponds slightly more closely to the direct measurement of voter preferences. Recall at 50% vote share, Democrats win 36% of the seats according to the direct measure, with an upper confidence bound at 44%. The weighted swing method yields 42.3% Democratic seat share, while the uniform swing method yields 47.3%.³⁰ For another example, consider the vote share required by Democrats to win 50% of the seats. For the direct

²⁷ For example, the 2002 dummy is a variable equal to one if the year is 2002 and zero otherwise. This controls for the year by subtracting out the average year value while controlling for other variables.

²⁸ Using state fixed effects instead of presidential vote share resulted in similar point estimates (with higher variance due to fewer effective degrees of freedom). The magnitude of the incumbency effect estimated here is similar to what was found in [18] for the Senate from 1982–2000.

²⁹ A caveat and a warning: we tried a mixed model which allowed for both fixed incumbency and time effects and interaction effects. However, there was massive multicollinearity between the fixed and interaction terms, resulting in high standard errors. The point estimates of the time interaction terms had the wrong sign. That is, more centrists resulted in less sensitivity to year effects. However, because of the high standard errors the combined model was not statistically different from the no direct effect model according to a Hausman test [19].

³⁰ The reason for the fractional seat share is the fact that each Missouri Senate race is weighted as only two-thirds.

measure, the point estimate is 51.9% with lower and upper confidence bounds of 50.7% and 53%. For the weighted swing method the vote share required is 51.4%; for the uniform swing method the vote share required is 50.3%.

The differences between the US Senate and US Presidential results deserve comment. Even though these elections span the same time frame (2002–2006) and the same constituencies (states), we find consistent evidence of partisan bias in one context but not the other. That difference reflects two facts. First, the Electoral College system apportions delegates on the basis of population as well as geography (state) and that the apportionment lessens the biases found in the distribution of state medians. Second, the US Senate and US President have different constituencies, owing in part to who votes in these elections and to the staggered timing of Senate elections.

5. Conclusions

This paper has considered a widely used empirical summary of elections, the relationship between party seat share and party vote share, in light of a spatial model of voting. The key insight derived from the model is that the seats–votes relationship is affected by two distributions, the distribution of voters and the distribution of district (or state) medians, and the standard analysis may conflate both distributions.

The one-dimensional spatial model underlies a new approach for quantitatively estimating the empirical relationship between seats and votes and helps in giving a more refined interpretation of the statistical results. In particular, we have shown how to estimate the distribution of district medians H . Specifically, national-level polling can allow estimation of the distribution of voters' ideal points, which in turn permits the use of aggregate election returns to estimate the distribution of district medians, H . This distribution, in turn, can be used to analyze how district composition affects the shares of seats won across hypothetical elections.

Finally, we have proposed several techniques, based on the model, that allow alternate approaches to construct the seats–votes relationship using data from only one election year, such as the weighted swing method and direct measurement of district medians. These methods have several advantages over regression methods, and in the examples considered here they yield fairly consistent results.

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